

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

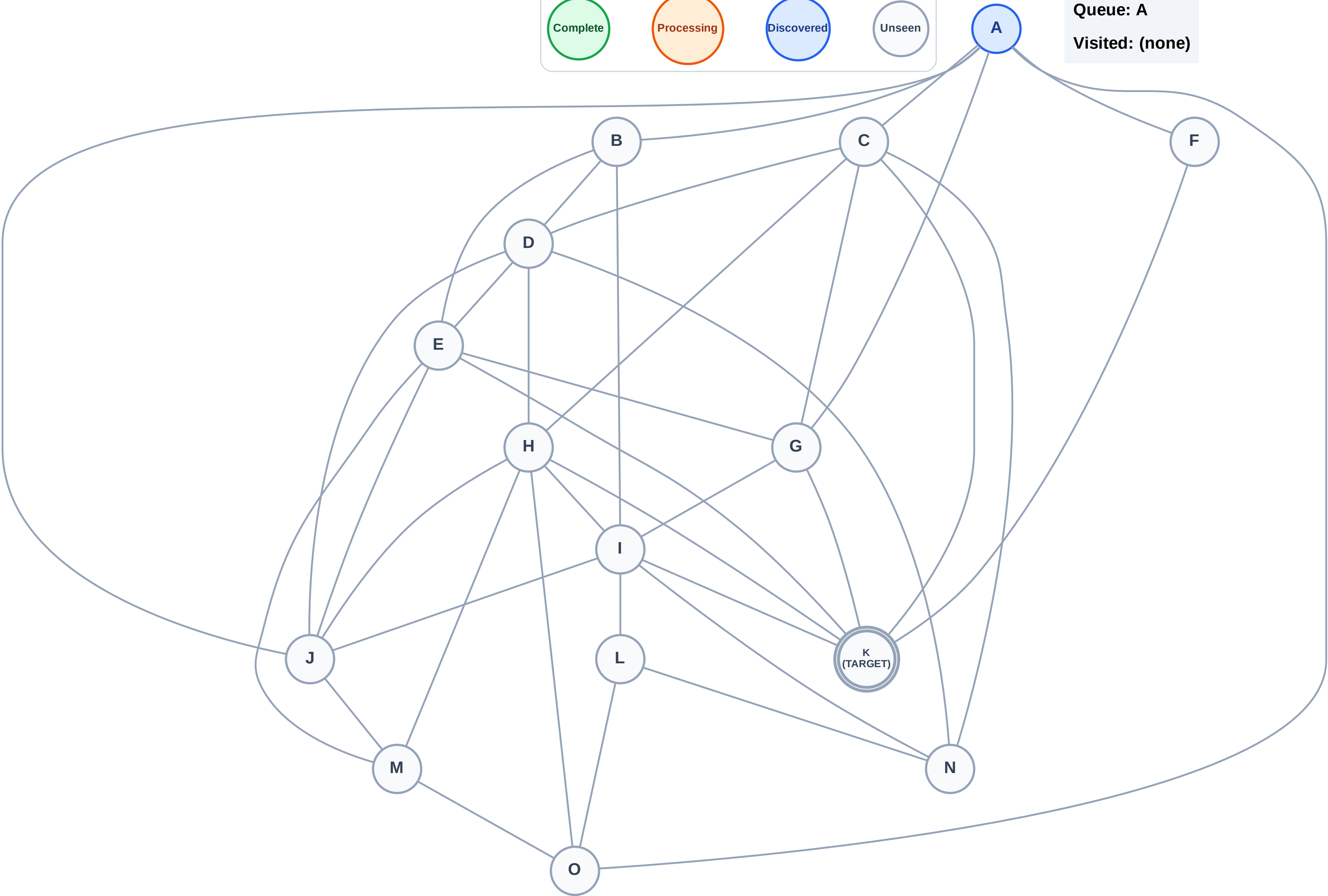
Complete

Processing

Discovered

Unseen

Queue: A
Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

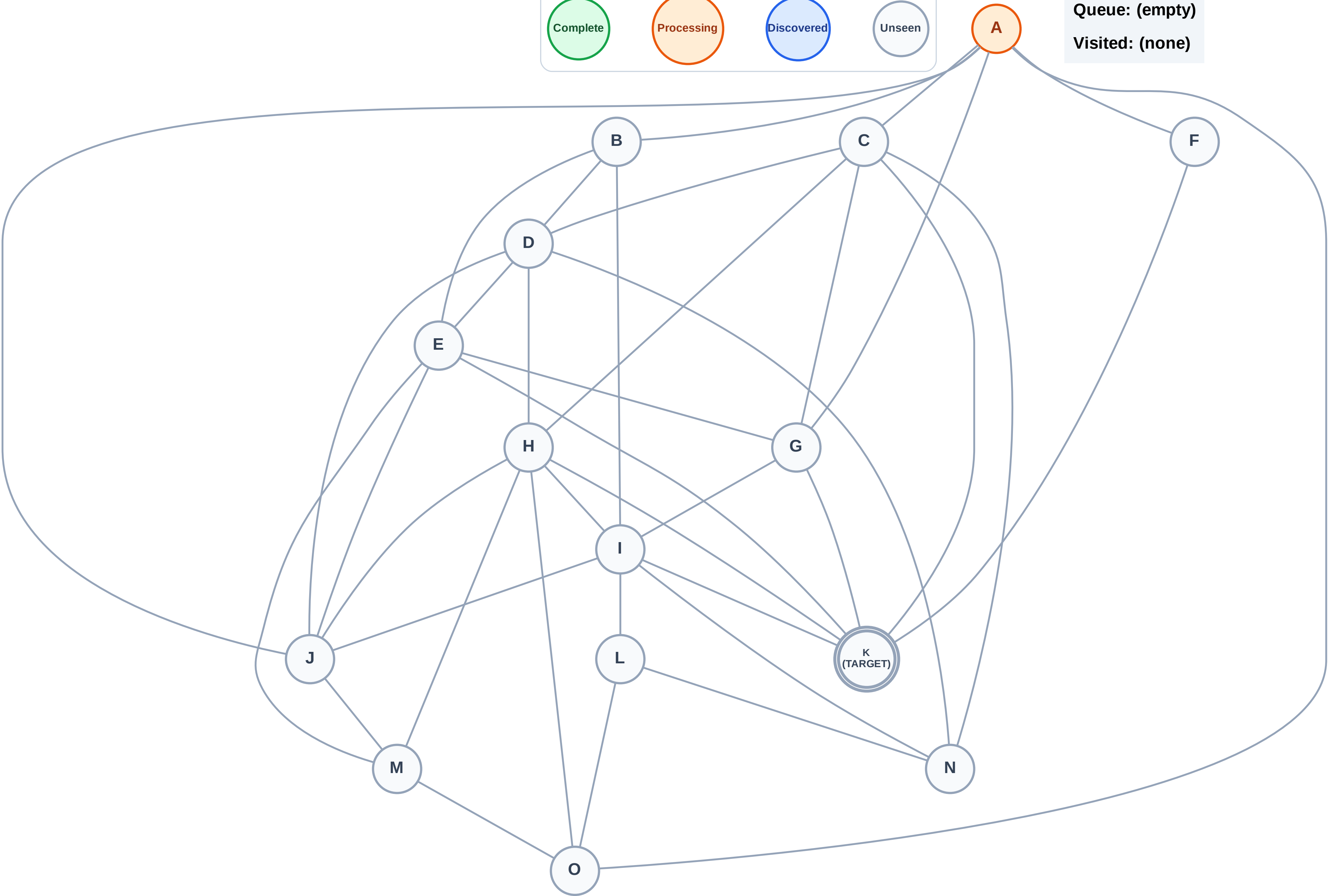
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

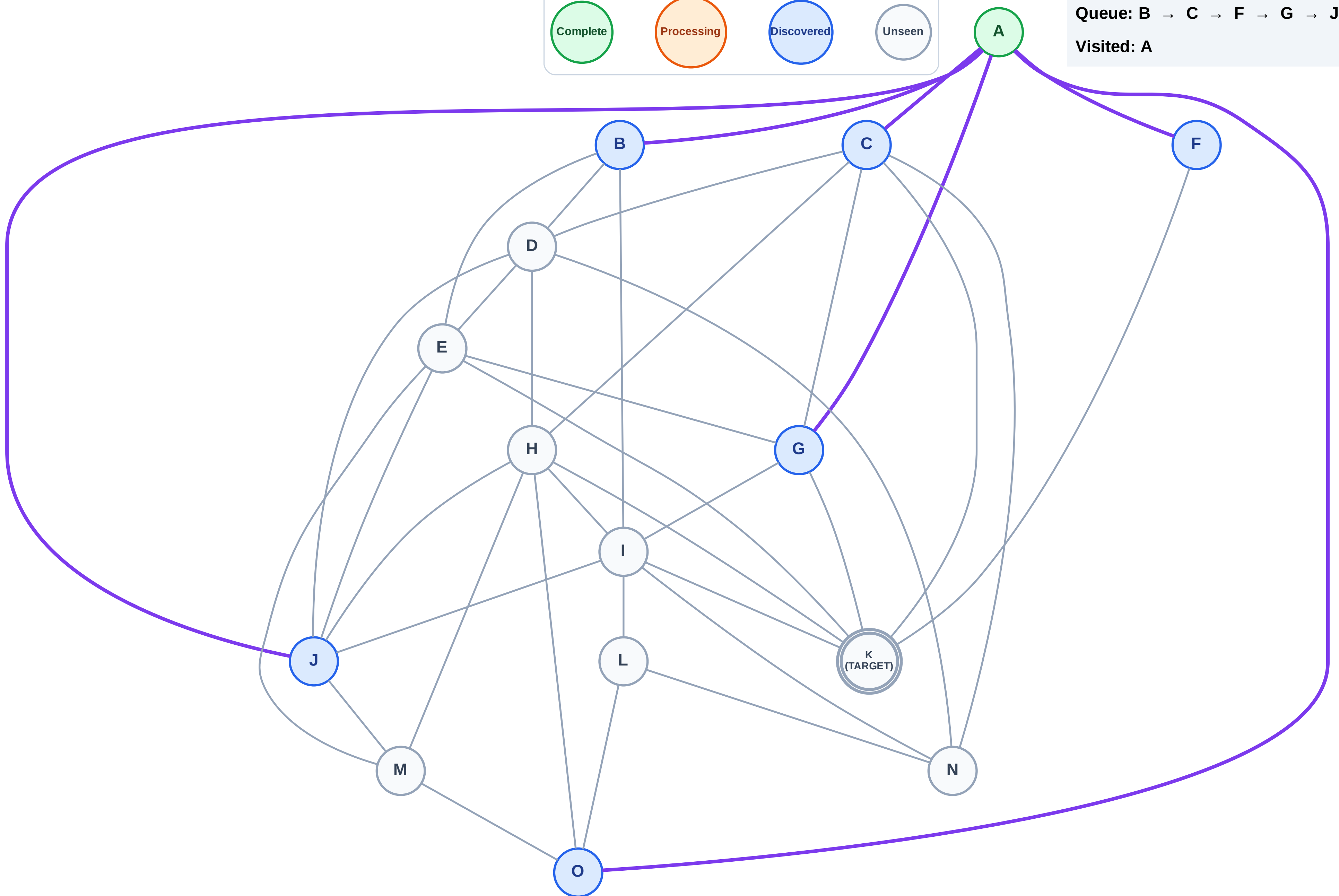
Step 3: Discover B, C, F, G, J, O from A

Legend



Queue: B → C → F → G → J → O

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

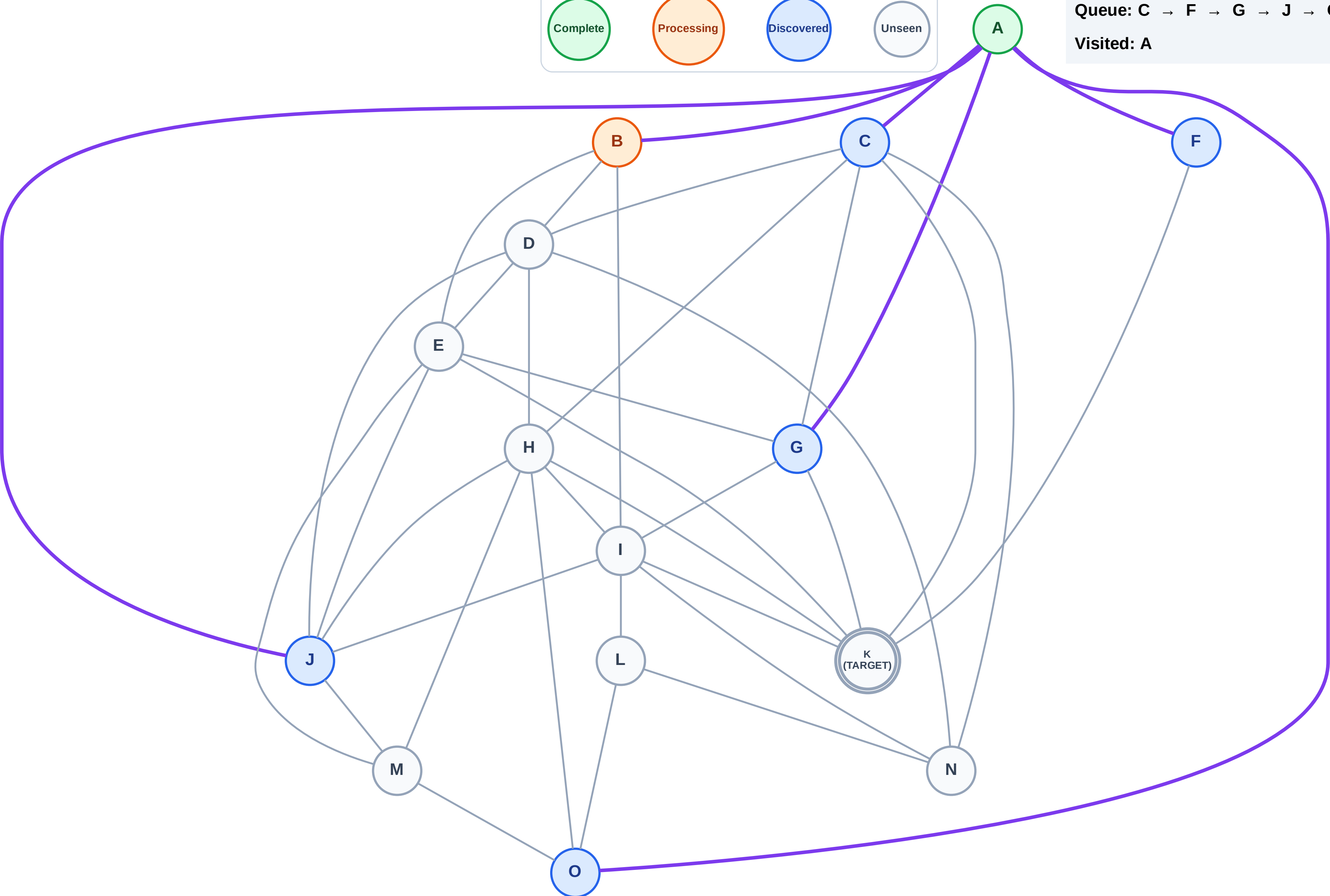
Processing

Discovered

Unseen

Queue: C → F → G → J → O

Visited: A



BFS Traversal

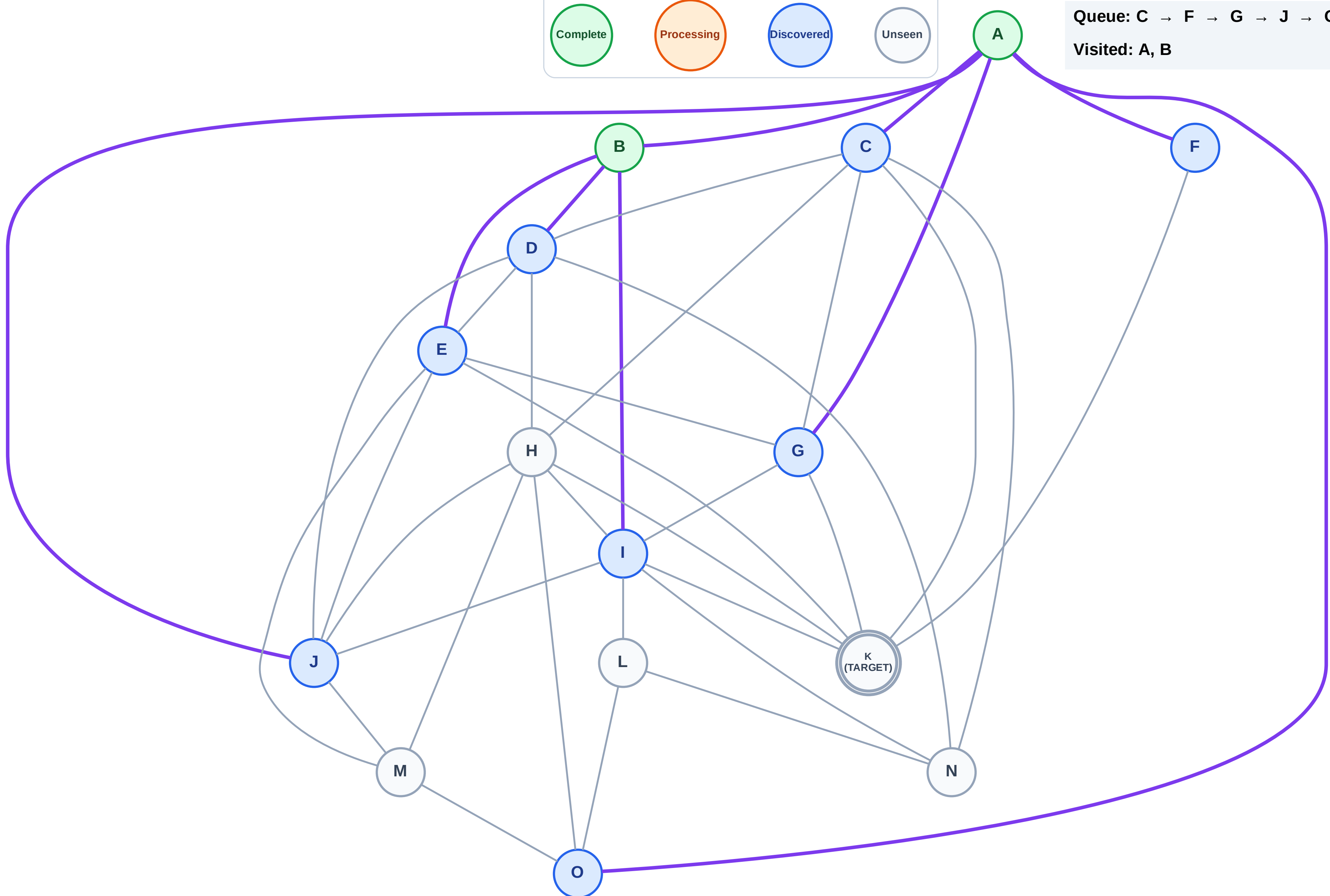
Step 5: Discover D, E, I from B

Legend



Queue: C → F → G → J → O → D → E → I

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

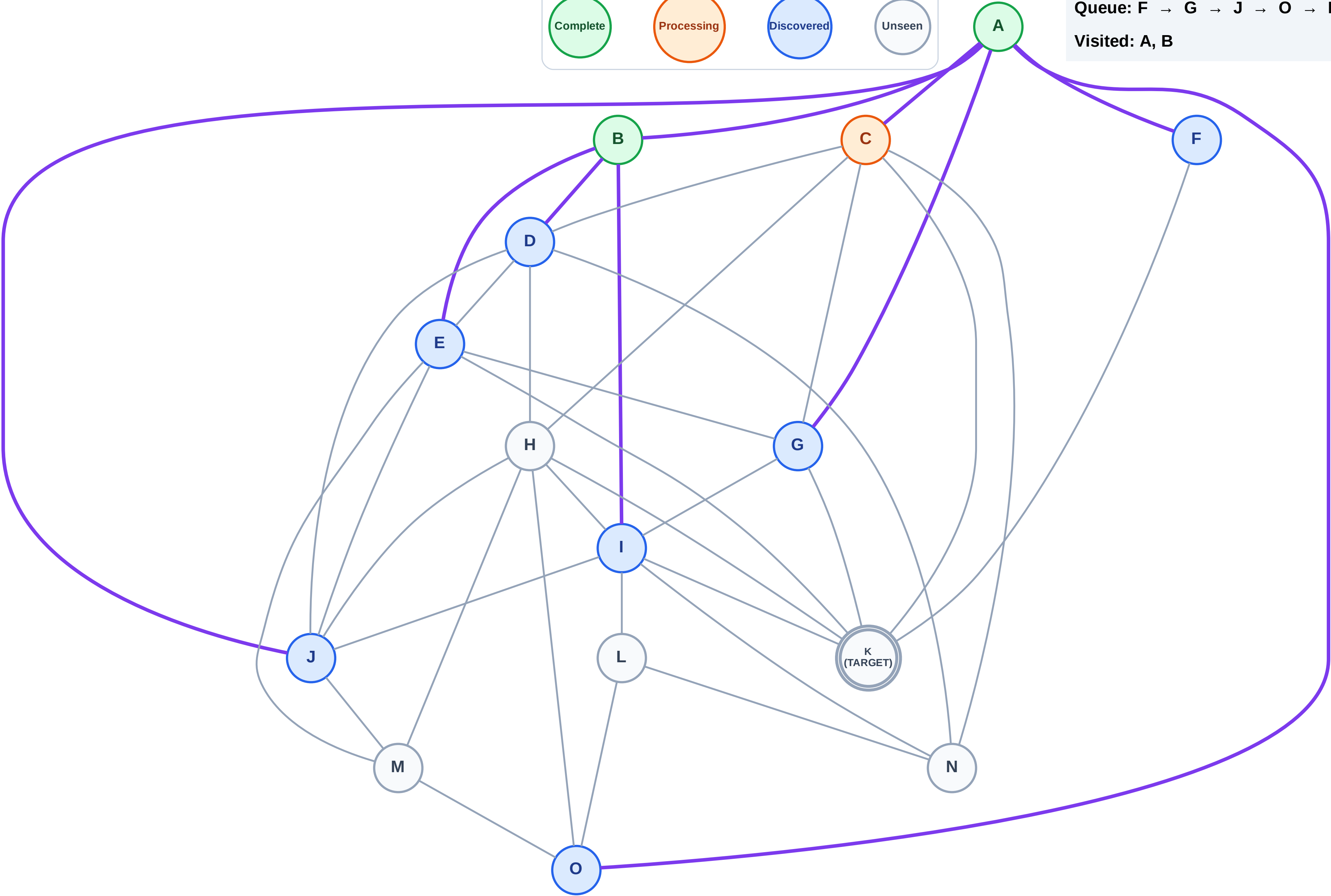
Processing

Discovered

Unseen

Queue: F → G → J → O → D → E → I

Visited: A, B



BFS Traversal

Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → C → K

Legend

Complete

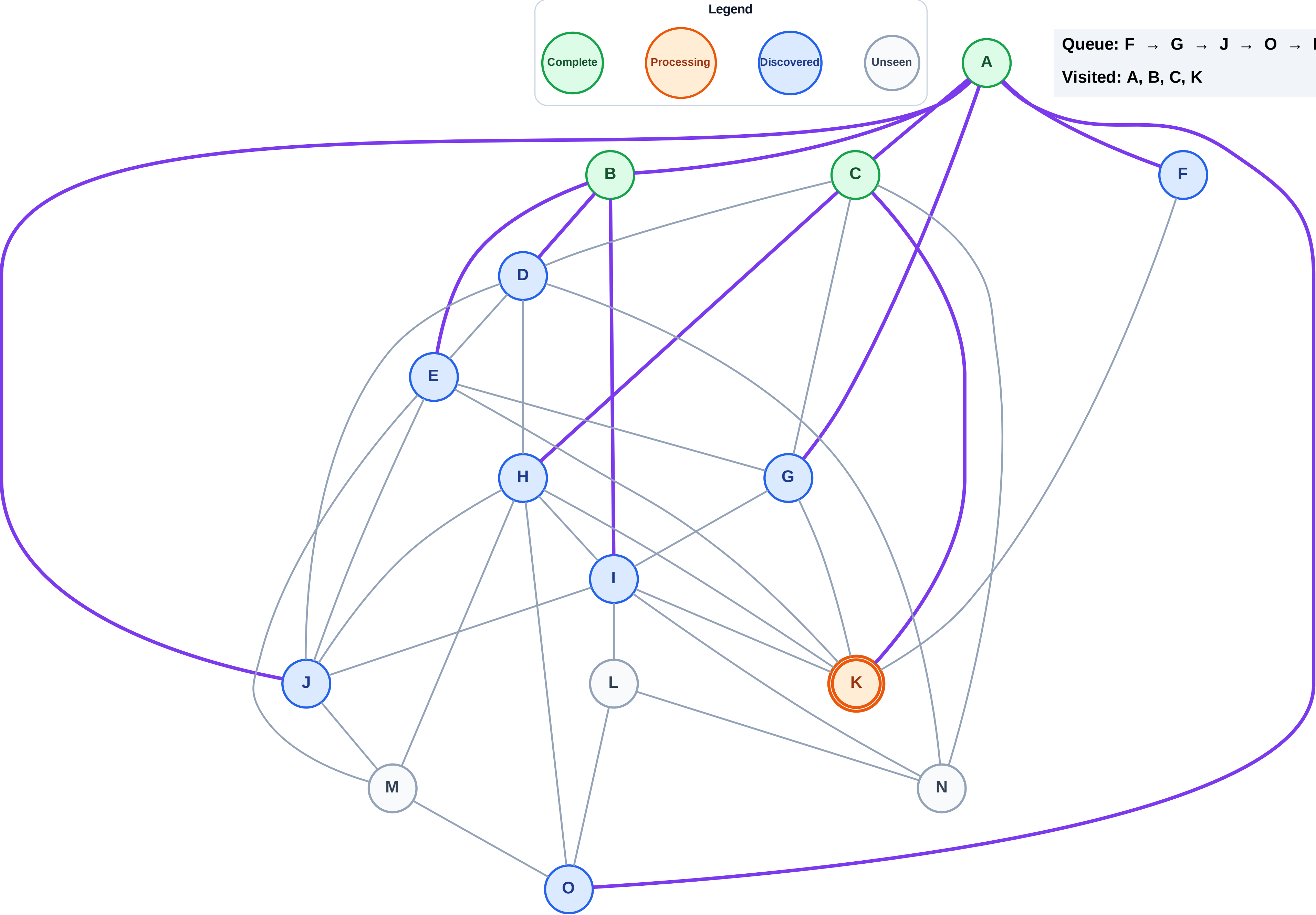
Processing

Discovered

Unseen

Queue: F → G → J → O → D → E → I → H

Visited: A, B, C, K



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

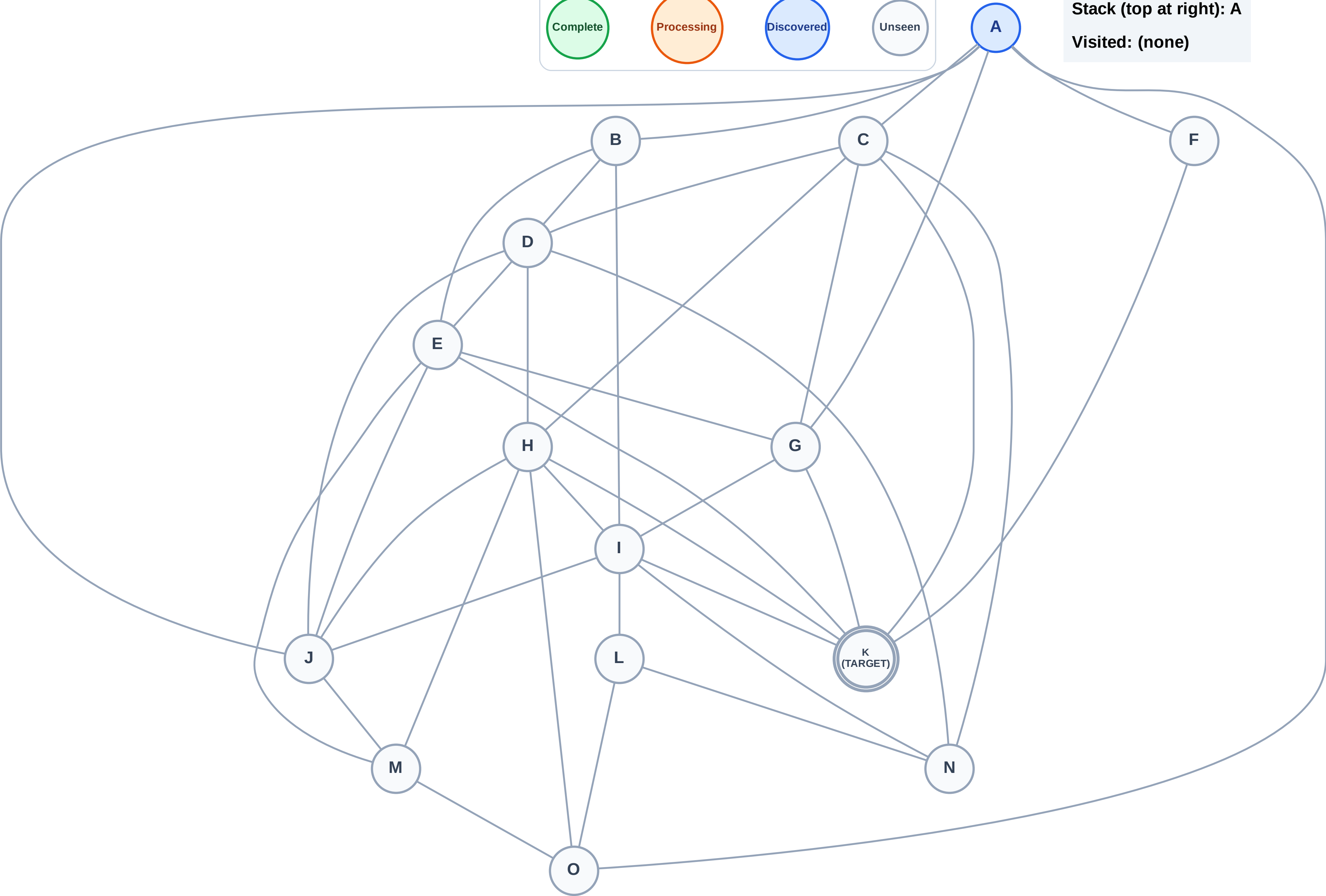
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

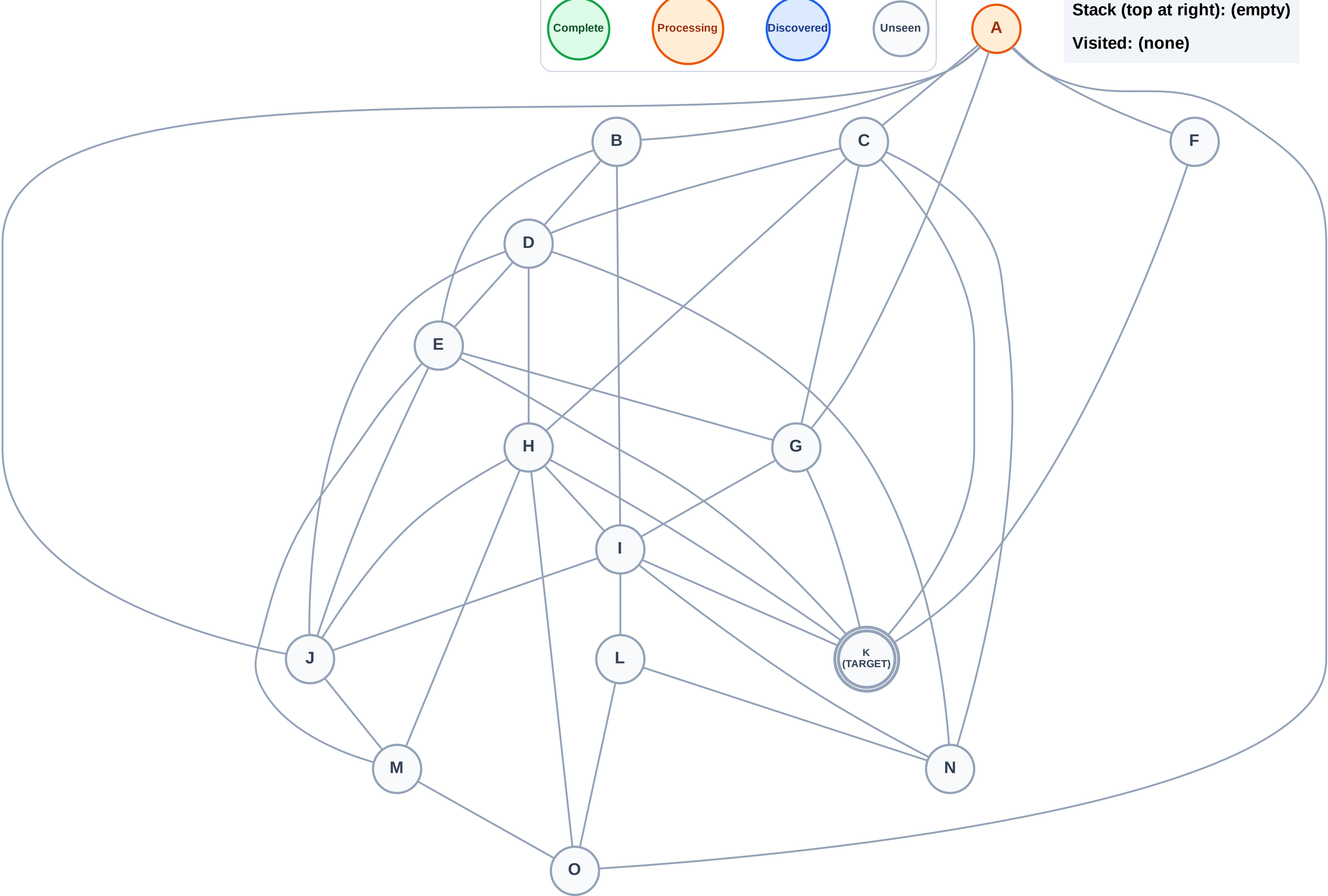
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

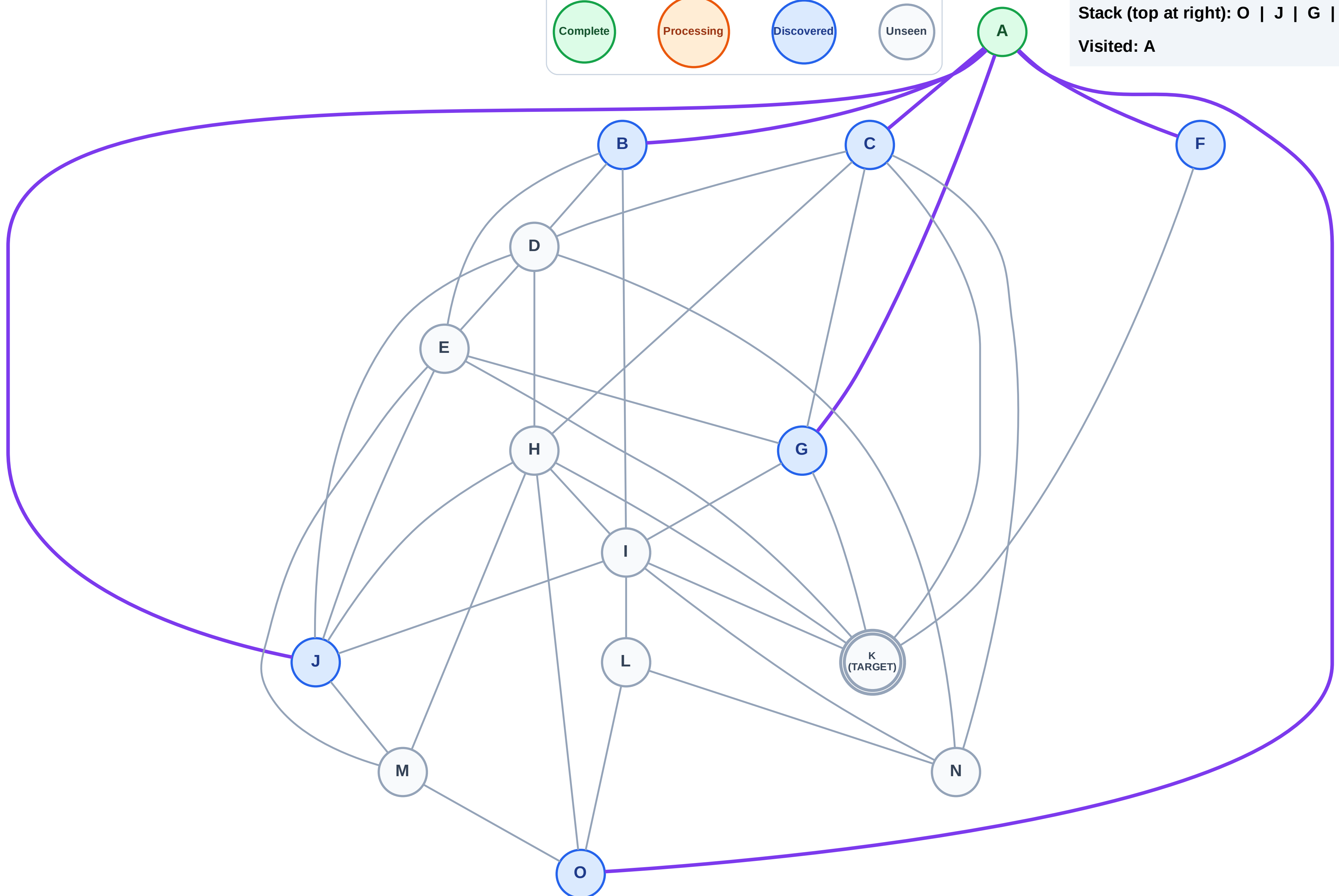
Step 3: Discover O, J, G, F, C, B from A

Legend



Stack (top at right): O | J | G | F | C | B

Visited: A



DFS Traversal

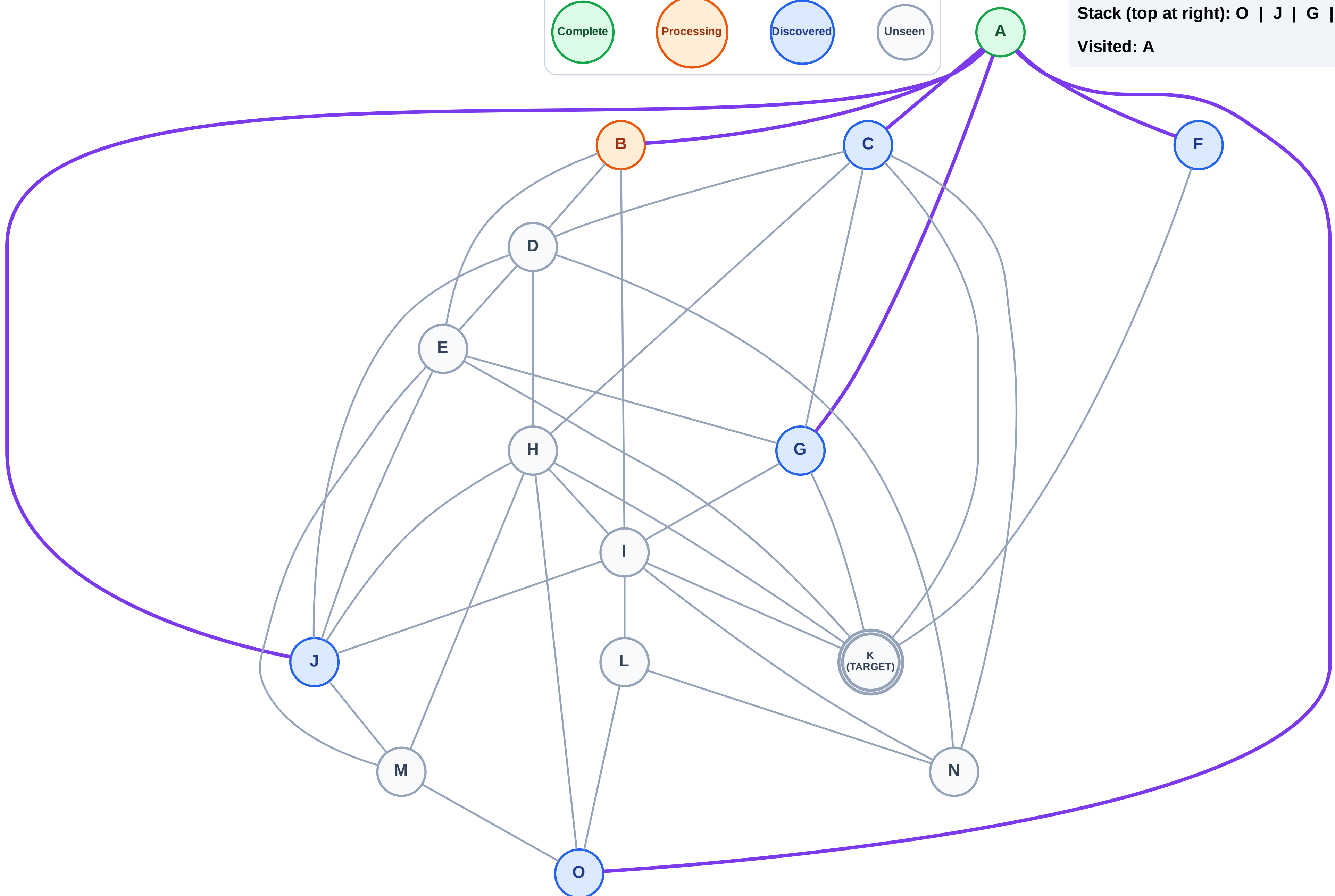
Step 4: Process node B

Legend



Stack (top at right): O | J | G | F | C

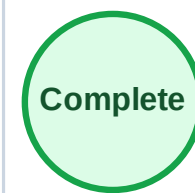
Visited: A



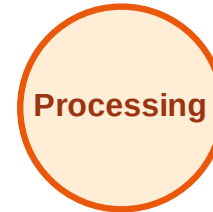
DFS Traversal

Step 5: Discover I, E, D from B

Legend



Complete



Processing



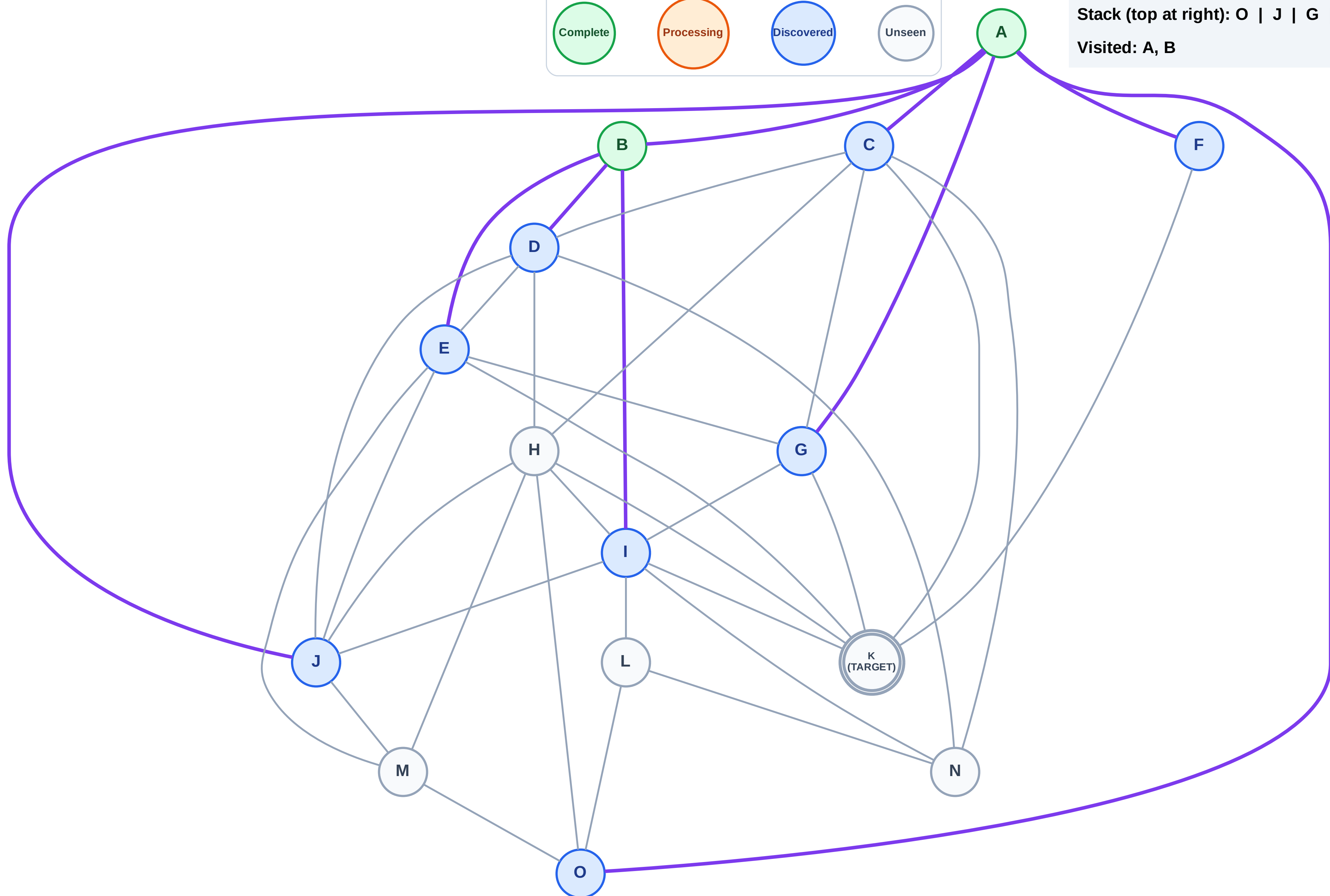
Discovered



Unseen

Stack (top at right): O | J | G | F | C | I | E | D

Visited: A, B



DFS Traversal

Step 6: Process node D

Legend

Complete

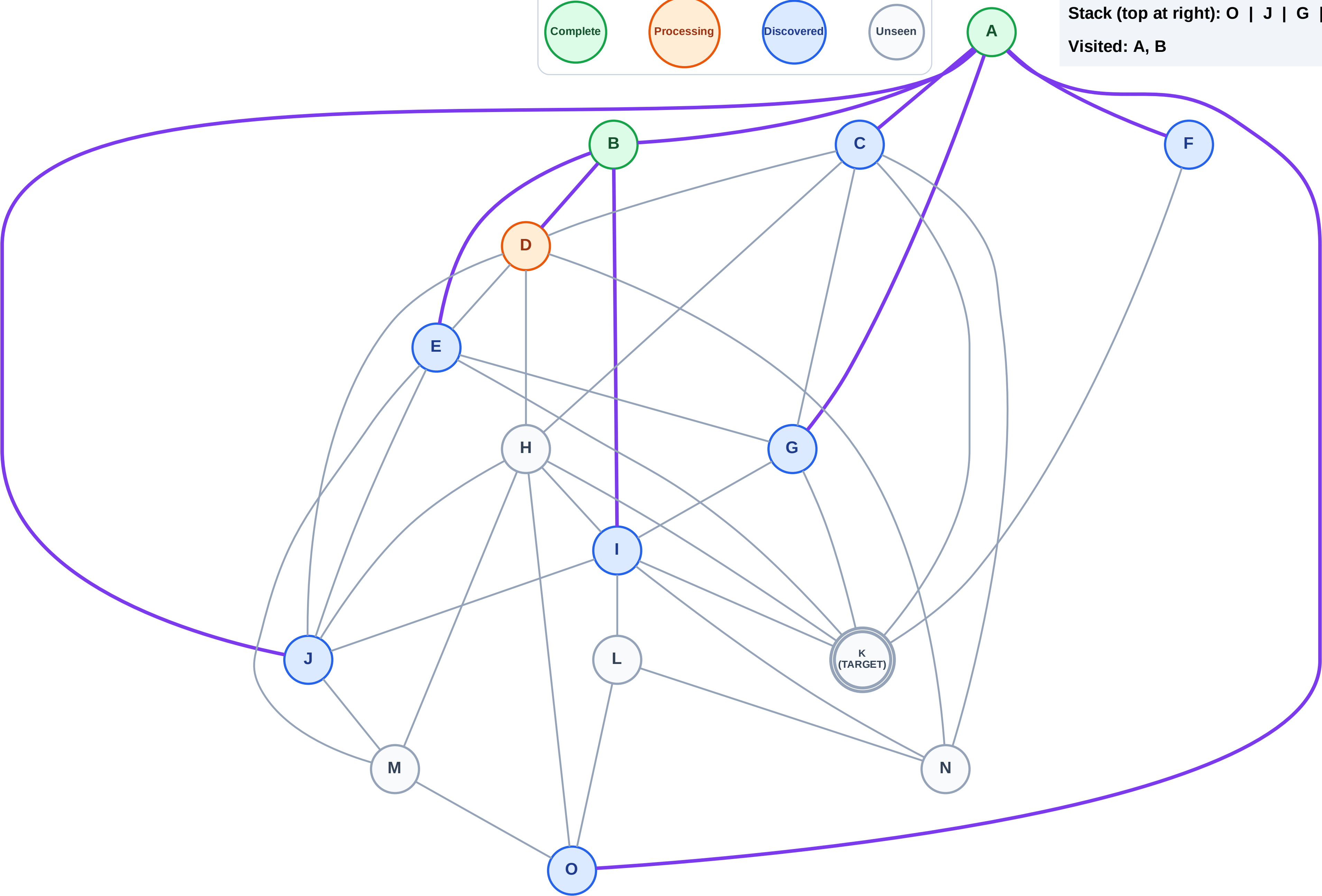
Processing

Discovered

Unseen

Stack (top at right): O | J | G | F | C | I | E

Visited: A, B



DFS Traversal

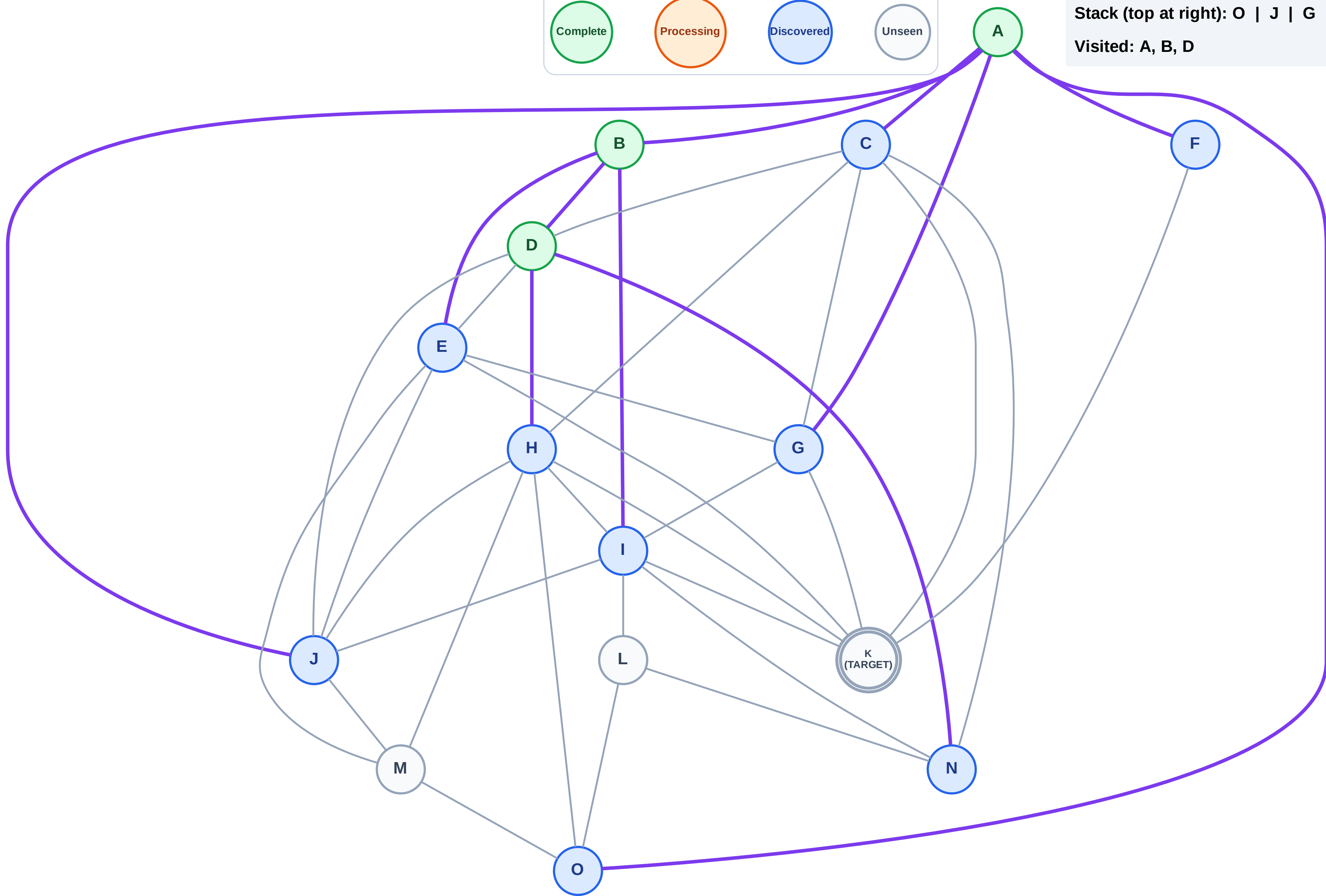
Step 7: Discover N, H from D

Legend



Stack (top at right): O | J | G | F | C | I | E | N | H

Visited: A, B, D



DFS Traversal

Step 8: Process node H

Legend

Complete

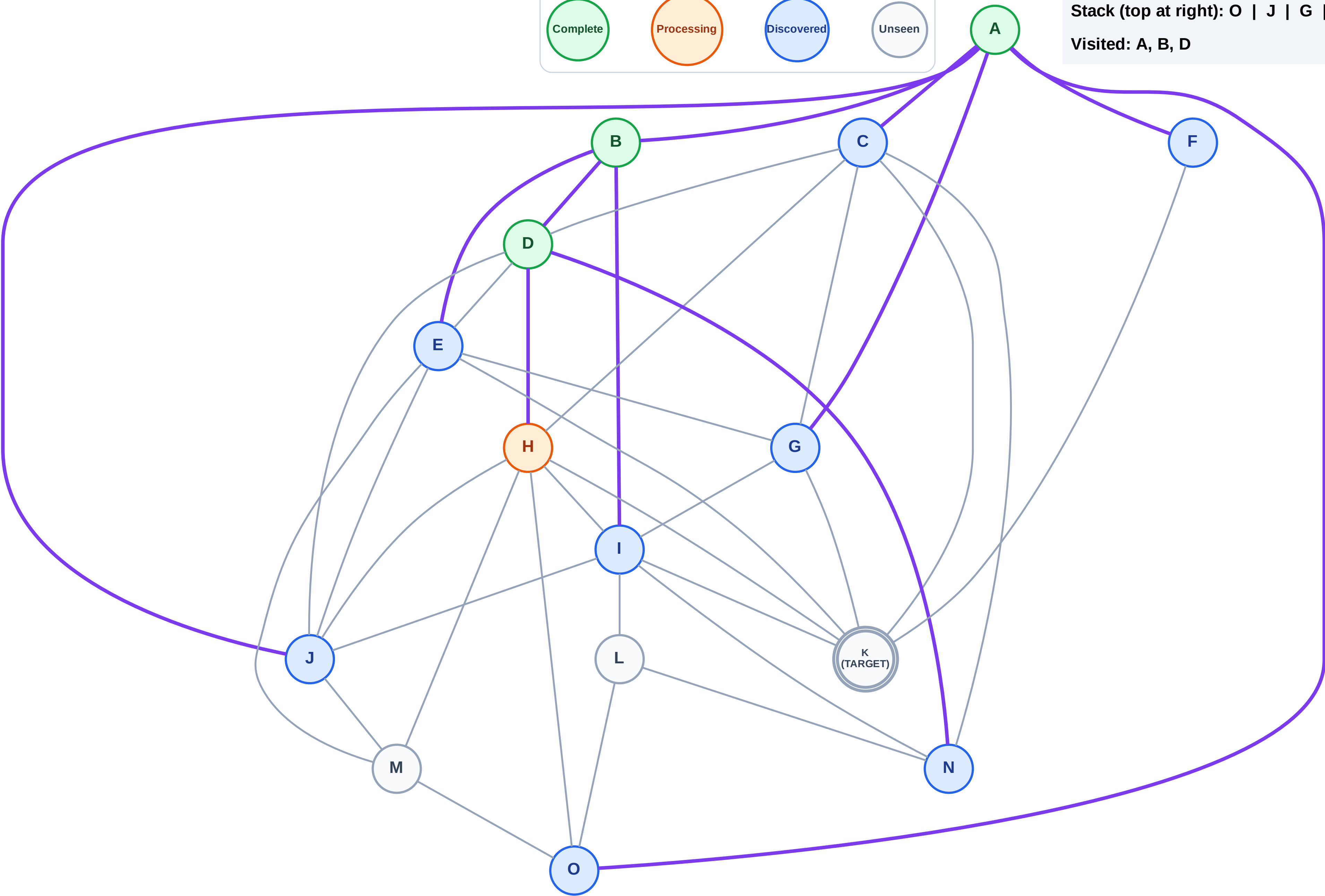
Processing

Discovered

Unseen

Stack (top at right): O | J | G | F | C | I | E | N

Visited: A, B, D



DFS Traversal

Step 9: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (4 hops): A → B → D → H → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): O | J | G | F | C | I | E | N | M

Visited: A, B, D, H, K

